Chapter 4 Breakout (Pt. II)

STAT 4714

Questions

1. Let X be a random variable with the following probability distribution:

x	-2	3	5
$f(x)$	0.3	0.2	0.5

   Find the standard deviation of X.

2. Suppose that the probabilities that 0, 1, 2, or 3 power failures will strike a certain subdivision in any given year are 0.4, 0.3, 0.2, and 0.1, respectively. Find the mean and variance of the random variable X representing the number of power failures striking this particular subdivision.

3. The length of time (in minutes) for a signal to flow through a communications channel is a random variable $Y = 3X - 2$, where X has the density function
$$f(x) = \frac{1}{4}e^{-x/4}, \quad x > 0.$$
   Find the mean and variance of the random variable Y.

1. Let X be a random variable with the following probability distribution:

x	-2	3	5
$f(x)$	0.3	0.2	0.5

Find the standard deviation of X.

$$\sigma_X^2 = E(X^2) - E(X)^2$$

$$E(X) = -2(0.3) + 3(0.2) + 5(0.5) = 2.5$$

$$E(X^2) = (-2)^2(0.3) + 3^2(0.2) + 5^2(0.5) = 15.5$$

$$\Rightarrow \sigma_X^2 = 15.5 - (2.5)^2 = 9.25$$

$$\Rightarrow \sigma_X = \sqrt{9.25} = \underline{3.04}$$

2. Suppose that the probabilities that 0, 1, 2, or 3 power failures will strike a certain subdivision in any given year are 0.4, 0.3, 0.2, and 0.1, respectively. Find the mean and variance of the random variable X representing the number of power failures striking this particular subdivision.

x	0	1	2	3
$P(X=x)$	0.4	0.3	0.2	0.1

$$E(X) = 0(0.4) + 1(0.3) + 2(0.2) + 3(0.1) = \underline{1}$$

$$E(X^2) = 0^2(0.4) + 1^2(0.3) + 2^2(0.2) + 3^2(0.1) = 2$$

$$\sigma_X^2 = E(X^2) - E(X)^2 = 2 - 1^2 = \underline{1}$$

3. The length of time (in minutes) for a signal to flow through a communications channel is a random variable $Y = 3X - 2$, where X has the density function

$$f(x) = \frac{1}{4}e^{-x/4}, \quad x > 0.$$

Find the mean and variance of the random variable Y.

$$E(Y) = E(3X-2) = 3E(X) - 2$$

$$\sigma_Y^2 = \sigma_{3X-2}^2 = 9\sigma_X^2 = 9\left[E(X^2) - E(X)^2\right]$$

$$E(X) = \int_0^\infty x \cdot \frac{1}{4} e^{-x/4}\,dx = \frac{1}{4}\int_0^\infty x e^{-x/4}\,dx$$

$u = x \qquad v = -4e^{-x/4}$

$du = dx \qquad dv = e^{-x/4}dx$

$$= \frac{1}{4}\left[-4xe^{-x/4}\Big|_0^\infty + 4\int_0^\infty e^{-x/4}\,dx\right]$$

$$= \int_0^\infty e^{-x/4}\,dx = -4e^{-x/4}\Big|_0^\infty = 4$$

$$\Rightarrow \underline{E(Y) = 3(4) - 2 = 10}$$

2 integration by parts!

$$E(X^2) = \int_0^\infty x^2 \cdot \frac{1}{4} e^{-x/4}\,dx = \ldots = 32$$

$$\Rightarrow \underline{\sigma_Y^2 = 9\left[32 - 4^2\right] = 144}$$